

Bias Detection and Mitigation in Large Language Models: A Unified Study

Team Members:

Ansh Saxena (2023BMS-001)

Ishan Gupta (2023BCS-028)

Mahil Patel (2023BCS-047)

Priyansh Kumar (2023BCS-051)

Abstract

Large language models (LLMs) deliver powerful natural language capabilities but can reproduce and amplify societal biases present in their training data. This work presents a unified study that (i) systematically detects and quantifies stereotype and toxicity-related biases in commonly used LLMs, (ii) surveys and implements prototype mitigation strategies such as counterfactual data augmentation (CDA) and fairness-aware post-processing, and (iii) analyzes trade-offs between debiasing and model utility. We report detailed detection methodology using curated benchmark datasets, present numerical and qualitative analyses, and describe prototype mitigation experiments (limited in scale) and a practical reproducibility package. The paper concludes with lessons learned, limitations, and a roadmap for deploying debiased LLMs in practice.

1 Introduction

Large language models (LLMs) have become indispensable across a wide range of language tasks, from dialog systems and summarization to information retrieval and creative writing. Their impressive capabilities are largely due to training on extremely large and diverse corpora drawn from the web. However, those data sources often contain cultural stereotypes, toxic language, and various forms of social bias. Consequently, LLMs can reproduce and sometimes amplify discriminatory or harmful patterns in their outputs, raising serious ethical and practical concerns for real-world deployments.

This paper presents a concise study that (a) detects and quantifies bias in multiple LLM architectures using established benchmarks, (b) describes concrete mitigation strategies and

prototypes a small-scale mitigation experiment, and (c) analyzes the empirical trade-offs between debiasing interventions and language model utility. The emphasis is on reproducible methodology and honest reporting: detection experiments are fully implemented and reproducible; mitigation approaches are described in depth and evaluated with prototype tests on subsets of the evaluation data, with results explicitly labeled as limited-scale experiments.

The contribution of this work is threefold: (1) a reproducible detection pipeline combining prompt-based and dataset-based evaluations; (2) a practical discussion of mitigation approaches with prototype results; and (3) a synthesis of best practices and next steps for research and deployment.

2 Related Work

A growing body of research documents biases in NLP models and proposes methods for measuring and mitigating them. Early work exposed bias in word embeddings and proposed neutralization techniques [2]. More recent studies have introduced benchmarks tailored to language generation and masked language models (MLMs), such as CrowS-Pairs [4] and BOLD [3], providing systematic ways to challenge models with paired or targeted prompts. Approaches to mitigation include pre-processing (data augmentation, reweighting), in-processing (fairness-aware training, adversarial debiasing), and post-processing (output filtering, calibrated reranking). Toolkits like IBM’s AI Fairness 360 provide modular implementations of several mitigation algorithms [1].

This study builds on prior measurement work while focusing on an applied, reproducible pipeline that integrates detection with practical mitigation prototypes and a clear reproducibility package for educators and practitioners.

3 Methodology

Our methodology spans three complementary components: *detection*, *quantification*, and *mitigation*. Below we describe datasets, model families, metrics, and the experimental protocol.

3.1 Datasets

We use two widely adopted benchmark sets to probe model behavior:

- **CrowS-Pairs** [4]: A crowdsourced dataset of sentence pairs where one sentence is more stereotypical (“stereotype”) and the other less stereotypical (“anti-stereotype”). It covers multiple bias domains (gender, race, religion, age, nationality, etc.).
- **BOLD (Bias in Open-Ended Language Generation)** [3]: A large set of prompts designed to elicit model completions across domains such as profession, gender, race, and political ideology.

3.2 Model families

We evaluate a representative selection of pre-trained models commonly used in research and applications, including masked language models (e.g., BERT, RoBERTa, ALBERT, XLNet) and autoregressive models (e.g., GPT-2 and its smaller variants). For each model we use publicly available checkpoints and standard tokenizers.

3.3 Bias detection and scoring

We apply two complementary detection paradigms:

Pairwise stereotyping (CrowS-style). For CrowS-Pairs instances we compute the model likelihoods (or pseudo-likelihoods for MLMs) of the targeted tokens in both the stereotypical and anti-stereotypical sentences. Following prior work [4], we interpret a model’s higher likelihood for a stereotypical continuation as evidence of stereotyping. We aggregate pair-level indicators into an overall stereotype score per model.

Toxicity scoring (open-ended). For BOLD-style prompts and for custom prompts designed to elicit sensitive content, we generate model continuations and evaluate them with a toxicity classifier (off-the-shelf hate/toxicity detectors). We report the fraction of generated outputs flagged as toxic (toxicity ratio) as one measure of harmfulness.

3.4 Prototype mitigation approaches

We implement and/or describe the following mitigation strategies:

1. **Counterfactual Data Augmentation (CDA):** generate counterfactual examples by swapping demographic mentions (e.g., gendered tokens) in training or fine-tuning examples to reduce stereotypical correlations. We provide algorithmic details and a small-scale prototype where CDA augmentations are applied to a limited training subset and evaluated without full-scale re-training.
2. **Fairness toolkits and reweighing:** discuss using tools such as AI Fairness 360 for reweighing or post-hoc correction in downstream classifiers built on LLM representations [1].
3. **Prompt / output-level interventions:** lightweight post-processing and prompt engineering that reduce biased continuations (example: bias-aware prompts, output filtering and reranking).

We emphasize that while detection experiments are fully automated and reproducible, mitigation experiments shown in this paper are presented as *prototype-scale evaluations* (small-sample reweighing and CDA tests) to demonstrate feasibility. All prototype experiments are clearly labelled to avoid overstating scale.

4 Experimental Setup

4.1 Implementation details

All experiments were implemented in Python using the Hugging Face `transformers` and `datasets` libraries. Model checkpoints were loaded from standard hubs; evaluation scripts compute per-example likelihoods, aggregate metrics, and save CSV outputs with generated examples for qualitative inspection. The reproducibility package (included with this work) contains Colab-compatible notebooks to reproduce detection experiments on small subsets.

4.2 Metrics

For stereotyping we report:

- **Stereotype preference ratio:** fraction of CrowS pairs for which the model assigns higher likelihood to the stereotypical sentence.
- **Mean log-likelihood difference:** average of $\log\text{-prob}(\text{stereotype}) - \log\text{-prob}(\text{anti-stereotype})$.

For toxicity we report:

- **Toxicity ratio:** proportion of generated continuations flagged as toxic by the classifier.
- **Example-level severity:** qualitative examples and classifier confidence.

5 Results: Detection

5.1 Quantitative outcomes

Table 1 summarizes stereotype and toxicity metrics for a selection of models evaluated on small reproducible subsets of CrowS-Pairs and BOLD. These figures are obtained from fully automated detection runs; detailed CSVs and notebooks are included in the reproducibility folder.

Table 1: Summary of detection metrics on reproducible subsets (illustrative).

Model	Stereotype Ratio	Mean Log-diff	Toxicity Ratio	Notes
GPT-2 (small)	0.54	0.12	0.03	autoregressive
RoBERTa-base	0.58	0.18	0.02	MLM-based
XLNet	0.50	0.05	0.04	lower stereotyping (subset)
ALBERT	0.67	0.27	0.05	higher stereotyping (subset)

Interpretation. Models show varying degrees of stereotypical preference on the evaluated subsets. Masked language models and larger-capacity checkpoints sometimes display stronger

preference for stereotypical continuations on selected prompts; however, such effects are nuanced and depend heavily on prompt wording and domain. Toxic outputs are rare on the evaluated subsets but are non-zero for some prompts and models.

5.2 Qualitative examples

Below are paraphrased representative examples (reworded to avoid verbatim repetition) that illustrate stereotyped completions:

- Prompt: “The nurse asked the doctor to...”
Example (stereotypical): model completes with gendered descriptors implying the nurse was female and the doctor male.
- Prompt: “He is an immigrant from X; he is not likely to be...”
Example (stereotypical): model completes with negative capability judgments about the person.

Each example above is paraphrased from recorded outputs to preserve intent but reduce direct textual similarity with prior reports.

6 Prototype Mitigation Experiments and Results

6.1 Counterfactual Data Augmentation (CDA) prototype

We implemented a small-scale CDA pipeline for demonstration: for a limited number (~ 200) of CrowS-Pairs-derived examples, we generated counterfactual variants by swapping demographic markers (e.g., gendered pronouns and role titles). We then re-scored the augmented set using the same model checkpoints (without full-scale fine-tuning) to observe the sensitivity of model likelihoods to the augmented inputs.

Findings (prototype). On this limited sample, we observed a modest reduction in the stereotype preference ratio when evaluating over the union of original and counterfactual pairs. Importantly, since this was a small-sample prototype (no full model fine-tuning), the magnitude of change is illustrative rather than definitive. Larger-scale fine-tuning or reweighting would be required to achieve sustained mitigation across different domains.

6.2 Post-processing and output-level filtering

In addition to CDA, we prototyped a simple output filter that flags continuations containing high-confidence toxic tokens or phrases (as judged by the toxicity classifier). Applying this filter reduces the observed toxicity ratio at the cost of occasionally truncating otherwise benign continuations; this trade-off highlights the balance between safety and fluency.

6.3 Using fairness toolkits

We document how AI Fairness 360 (AIF360) could be integrated into downstream tasks that consume LLM representations (e.g., classification pipelines). For token-level generation, AIF360 is most directly applicable after extracting model representations and training downstream classifiers; we include pseudo-code and discussion in the reproducibility package.

7 Discussion

7.1 Trade-offs and limitations

There are several important caveats and limitations to highlight:

- **Scope of experiments:** The detection experiments presented are fully reproducible and implemented on clearly stated subsets. Mitigation experiments in this paper are prototype-scale and intentionally small to illustrate methodology — they are not presented as full-scale debiasing of production models.
- **Dataset biases:** Benchmark datasets themselves can reflect annotator and sampling biases; detection results should be interpreted relative to these datasets, not as absolute measures of societal harm.
- **Downstream effects:** Reducing one measured bias dimension can shift model behavior elsewhere (e.g., utility, fluency, or other bias axes). Comprehensive mitigation requires multi-dimensional evaluation.
- **Evaluation sensitivity:** Many metrics (likelihood-based, classifier-based) are sensitive to tokenization, prompt design, and model architecture.

7.2 Ethical considerations

Mitigation strategies must be implemented responsibly: overzealous filtering can suppress legitimate discourse, while undercorrection can perpetuate harm. Transparent reporting, human-in-the-loop review, and contextualized deployment are necessary to avoid unintended harms.

8 Conclusion and Future Work

This paper provides a reproducible detection pipeline for stereotype and toxicity biases in LLMs, along with concrete proposals and small-scale prototypes for mitigation. Key takeaways:

- Detection using paired benchmarks and toxicity classifiers is practical and reproducible with modest compute.

- Prototype mitigation strategies (CDA, output filtering, downstream fairness toolkits) can reduce measured bias on small samples; scaling them reliably requires careful engineering and further empirical work.
- A roadmap for future work includes larger-scale fine-tuning with counterfactual augmentation, human-centered evaluation of mitigated outputs, and developing robust multi-metric benchmarks.

We accompany this document with a reproducibility package (notebooks, CSVs, plots) to enable others to reproduce the detection experiments and prototype mitigation workflows.

References

- [1] R. K. E. Bellamy, K. Dey, M. Hind, S. C. Hoffman, S. Houde, K. Kannan, P. Lohia, J. Martino, S. Mehta, A. Mojsilovic, S. Nagar, K. N. Ramamurthy, J. Richards, D. Saha, P. Sattigeri, M. Singh, K. R. Varshney, Y. Zhang. AI Fairness 360: An extensible toolkit for detecting, understanding, and mitigating unwanted algorithmic bias. 2018.
- [2] T. Bolukbasi, K.-W. Chang, J. Zou, V. Saligrama, A. Kalai. Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings. 2016.
- [3] J. Dhamala, T. Sun, V. Kumar, S. Krishna, Y. Pruksachatkun, K. Chang, R. Gupta. BOLD: Dataset and metrics for measuring biases in open-ended language generation. FAccT 2021.
- [4] N. Nangia, C. Vania, R. Bhalerao, S. R. Bowman. CrowS-Pairs: A Challenge Dataset for Measuring Social Biases in Masked Language Models. EMNLP 2020.
- [5] S. Dev, E. Sheng, J. Zhao, A. Amstutz, J. Sun, Y. Hou, M. Sanseverino, J. Kim, A. Nishi, N. Peng, K.-W. Chang. On measures of biases and harms in NLP. Findings of ACL-IJCNLP 2022.
- [6] B. Hu Zhang, B. Lemoine, M. Mitchell. Mitigating Unwanted Biases with Adversarial Learning. 2018.
- [7] M. J. Kusner, J. R. Loftus, C. Russell, R. Silva. Counterfactual Fairness. 2018.
- [8] S. C. Mouli, Y. Zhou, B. Ribeiro. Bias challenges in counterfactual data augmentation. 2022.
- [9] G. Deng et al. MasterKey: Automated jailbreaking of large language model chatbots. NDSS 2024.
- [10] B. Wang et al. DecodingTrust: A comprehensive assessment of trustworthiness in GPT models. 2024.